RENGGLI PC SOLUTION - ENTERPRISE TOOLBOX V14

Complete Manual in English

Renggli PC Solution

2026-07-13

🖥️ RENGGLI PC SOLUTION - ENTERPRISE TOOLBOX V14

Multi-Platform Suite (Windows | Linux | macOS)

Professional diagnostic, repair and administration solution for IT technicians

🌐 CHOOSE YOUR OPERATING SYSTEM

This manual covers installation and usage on **Windows, Linux and macOS**.

Jump directly to your section: - 🪟 Instructions for WINDOWS - 🐧 Instructions for LINUX - 🍎 Instructions for macOS

🪟 WINDOWS

📋 System Requirements (Windows)

Requirement	Specification
Operating System	Windows 10/11 (Build 1809 or higher)
PowerShell	Version 5.1 or higher (pre-installed)
Privileges	Administrator (mandatory)
Disk Space	50 MB minimum

Requirement	Specification
Network	Internet connection (for updates)

⍰ Installation on Windows - STEP BY STEP

Step 1: Download the tool

1. Download the complete `Herramienta-toolbox` folder
2. Extract the ZIP file anywhere (Example: `C:\Tools\`)
3. You'll see this structure:

```
Herramienta-toolbox/
├── Windows/
│   └── toolbox.bat
├── Linux/
├── Mac/
└── Manuales/
```

Step 2: Navigate to Windows folder

1. Open **File Explorer** (Windows + E)
2. Navigate to where you extracted the tool
3. Enter the **`Windows`** folder
4. You'll see the file:
   - `toolbox.bat` (single Windows edition)

Step 3: Run as Administrator

⍰ **IMPORTANT**: You must run with administrator permissions

Method 1 - Right-click (Recommended):

1. **Right-click** on `toolbox.bat`
2. Select **"Run as administrator"**

```
┌─────────────────────────┐
│ Open                    │
│ Edit                    │
│ Print                   │
│ → Run as administrator  │ ← THIS OPTION
│ Share                   │
└─────────────────────────┘
```

3. If **User Account Control (UAC)** appears, click **"Yes"**

Method 2 - From CMD:

1. Open **CMD as administrator**:

- Press `Windows + X`
- Select “Terminal (Admin)” or “Command Prompt (Admin)”

2. Navigate to the folder:

```
cd C:\path\where\you\extracted\Herramienta-toolbox\Windows
```

3. Run:

```
toolbox.bat
```

Method 3 - From PowerShell:

1. Open **PowerShell as administrator**:
   - Press `Windows + X`
   - Select “Windows PowerShell (Admin)”

2. Navigate and run:

```
cd C:\path\where\you\extracted\Herramienta-toolbox\Windows
.\toolbox.bat
```

Step 4: Select Execution Profile

When you run it, you’ll see this screen:

```
==========================================================================================
                        Renggli PC Solution - SUITE ENTERPRISE V14
==========================================================================================
Current Log: C:\...\Logs\Audit_2024-02-10.log

[PROFILE SELECTION]

1. DIAGNOSTICS     - Read-only, audit and queries (no modifications)
2. REPAIR          - Automated maintenance and repairs
3. ADMINISTRATION - Full access (includes formatting and conversions)

=> Select profile [1-3]:
```

Choose your profile: - **1** = View information only, makes no changes - **2** = Allows repairs and cleanup - **3** = Full access (be careful with this)

Type the number and press **Enter**.

Step 5: Use the Main Menu

After selecting the profile, you’ll see the specific menu for that profile:

? If you chose DIAGNOSTICS (Profile 1):

```
==========================================================================================
   Active Profile: [DIAGNOSTICS] - Read Only

   This menu is query-only. It does not make system changes.
   Ideal for auditing hardware, resources, network, and Windows Update status.
   Note: battery report only applies to laptops.
   Legend: [R] Read-only  [W] Writes/changes system  [!] Critical/irreversible

   [ DIAGNOSTICS BASE ]
   1. [R] RAM Test (mdsched)
   2. [R] System Resources Info
   3. [R] BIOS and Motherboard Info
   4. [R] Windows Update Status
   5. [R] Ports/DNS Audit
   6. [R] Network Speed Test
   7. [R] Battery Report

   [ ADVANCED ANALYSIS - READ ONLY ]
   8. [R] Critical Events (System)
   9. [R] BSOD Analysis (Minidump)
   10. [R] Process Forensic Audit
   11. [R] RAID/Storage Status

   [0] EXIT WITH REPORT             [00] EXIT WITHOUT REPORT AND WITHOUT LOG
   [99] CHANGE PROFILE
==========================================================================================

=> Select an option:
```

This profile only has read-only options. It doesn't modify the system, only queries information.

� If you chose REPAIR (Profile 2):

```
=========================================================================================
   Active Profile: [REPAIR] - Maintenance and Repairs

   Includes diagnostics and tasks that modify the system.
   Recommended for maintenance, cleanup, and guided repairs.
   Note: battery report only applies to laptops.
   Legend: [R] Read-only  [W] Writes/changes system  [!] Critical/irreversible

   [ REPAIR AND MAINTENANCE ]
   1. [R] RAM Test (mdsched)
   2. [R] System Resources Info
   3. [R] BIOS and Motherboard Info
   4. [W] Maintenance (DISM/SFC)
   5. [W] Repair Windows Update
   6. [W] EMMC/Temp Cleanup
   7. [W] Network & IP Reset
   8. [R] Speed Test
   9. [R] Ports/DNS Audit
   10. [W] Scheduled Shutdown
   11. [W] Update Apps (Winget)
   12. [R] Battery Report
   13. [W] Driver Backup (WRITES TO DISK)

   [ ADVANCED ANALYSIS - READ ONLY ]
   14. [R] Critical Events
   15. [R] BSOD Analysis
   16. [R] Process Forensic Audit
   17. [R] RAID/Storage Status

   [0] EXIT WITH REPORT             [00] EXIT WITHOUT REPORT AND WITHOUT LOG
   [99] CHANGE PROFILE
=========================================================================================

=> Select an option:
```

This profile includes diagnostics + system repairs. Can perform maintenance but not critical operations.

⚠️ If you chose ADMINISTRATION (Profile 3):

```
================================================================================
   Active Profile: [ADMINISTRATION] - Full Access

   Full access. Includes irreversible actions and critical changes.
   Use this profile only if you understand each operation's impact.
   Note: battery report only applies to laptops.
   Legend: [R] Read-only  [W] Writes/changes system  [!] Critical/irreversible

   [ ADMINISTRATION OPERATIONS ]
   1. [R] BIOS and Motherboard Info
   2. [R] RAM Test (mdsched)
   3. [R] System Resources Info
   4. [W] Maintenance (DISM/SFC)
   5. [W] Repair Windows Update
   6. [W] EMMC/Temp Cleanup
   7. [W] Network & IP Reset
   8. [R] Real Speed Test
   9. [R] DNS/Ports Audit
   10. [!] Secure Format (Audited)
   11. [!] MBR to GPT Conversion
   12. [W] Update Apps (Winget)
   13. [W] MASTER ACTIVATION (MAS)
   14. [W] Scheduled Shutdown
   15. [R] Battery Report
   16. [W] Driver Backup (WRITES TO DISK)

   [ ADVANCED ANALYSIS - READ ONLY ]
   17. [R] Critical Events
   18. [R] BSOD Analysis
   19. [R] Process Forensic Audit
   20. [R] RAID/Storage Status
   21. [W] High Security Profile (Blindaje V1 integrated)

   [0] EXIT WITH REPORT             [00] EXIT WITHOUT REPORT AND WITHOUT LOG
   [99] CHANGE PROFILE
================================================================================

=> Select an option:
```

This profile has full access, including critical formatting/conversion tasks and option 21 for Blindaje V1.

To use a function: 1. Type the **option number** you want to use 2. Press **Enter** 3. Follow the on-screen instructions 4. The tool will guide you step by step

To change profile: - Type **99** and press Enter at any time - You can choose a different profile without restarting the tool

CLI Execution (optional)

You can also start the tool with parameters to preselect profile and module:

```
toolbox.bat /perfil:X /mod:Y
```

- `X` = profile (`1` Diagnostics, `2` Repair, `3` Administration)
- `Y` = module number inside that profile

Quick examples (full version `toolbox.bat`):

- Profile 1 (Diagnostics):
  - `toolbox.bat /perfil:1 /mod:1` (RAM test)

    - `toolbox.bat /perfil:1 /mod:4` (Windows Update status)
    - `toolbox.bat /perfil:1 /mod:11` (RAID/Storage status)
    - `toolbox.bat /perfil:1 /mod:12` (Disk SMART status)
- Profile 2 (Repair):
    - `toolbox.bat /perfil:2 /mod:4` (DISM/SFC)
    - `toolbox.bat /perfil:2 /mod:5` (Repair Windows Update)
    - `toolbox.bat /perfil:2 /mod:10` (Scheduled shutdown)
- Profile 3 (Administration):
    - `toolbox.bat /perfil:3 /mod:10` (Secure format)
    - `toolbox.bat /perfil:3 /mod:11` (MBR to GPT conversion)
    - `toolbox.bat /perfil:3 /mod:21` (High Security Profile)
    - `toolbox.bat /perfil:3 /mod:22` (PostgreSQL Password Manager)

Notes: - If `profile/module` is invalid for that profile, execution is blocked and logged. - Modules keep their safety confirmations before running critical actions.

Step 6: Exit and View Logs

To exit: - Type **0** to generate HTML report and exit - Type **00** to exit without generating report (log + checksum are saved)

About the HTML report: - Option **0** automatically generates an HTML file in the `Logs/` folder - The report includes all system information and operation logs - It will open automatically in your browser

---

? Where are the logs?

Logs are automatically saved in:

```
Windows/Logs/
├── Audit_2024-02-10.log         (Record of all operations)
├── Report_2024-02-10.html       (Visual report)
└── battery-report.html          (If you used battery function)
```

?? Troubleshooting (Windows)

Error: “You don’t have administrator privileges”

Solution: You must run with right-click → “Run as administrator”

Error: “The system cannot execute the script”

Solution: 1. Windows may have blocked the file 2. Right-click on `toolbox.bat` → Properties 3. If you see “This file came from another computer”, check “Unblock” 4. Click Apply → OK

The window closes immediately

Solution: Run from CMD or PowerShell to see the error

Color menu doesn't appear

Solution: Use Windows Terminal or CMD (not PowerShell ISE)

`bash -n` validation fails on Windows

Cause: `bash.exe` depends on WSL plus an installed/active Linux distribution.

Meaning: if WSL has no configured distro, `bash -n` fails even if the `.sh` script is valid.

Quick fix: 1. Install/enable WSL and a distro (e.g., Ubuntu). 2. Re-run syntax checks from Linux/WSL: - `bash -n Linux/toolbox.sh` - `bash -n Mac/toolbox.sh`

Classroom temp-file workflow (Option 21)

Inside **Profile 3 (Administration)**, **Option 21 (High Security Profile)** now includes:

- **Two apply paths** inside the same menu.
- **Strict hardening**: maximum protection for academic files/folders, but Office/Adobe style atomic save workflows can fail in academic folders; editing an existing file and saving over it can also fail.
- **Soft lock**: protects the academic folder structure detected under `%BL_ROOT_DIR%` (excluding `PERFIL` and `BGInfo`) while allowing normal save/delete of individual files.
- **Robust daily-folder redirection**: Desktop/Documents/Downloads/Music/Pictures/Videos now point to the physical `%BL_ROOT_DIR%\PERFIL\Usuario` path so slow Windows 10 machines do not fail when `T:` is not ready yet at sign-in.
- safe manual temp review/cleanup in `Trabajos Alumnos\SECUNDARIA` and `Trabajos Alumnos\PRIMARIA`
- local daily scheduled auto-clean task setup
- local auto-clean task removal
- mass deployment guide (domain/GPO and non-domain remote rollout)

Safe temp patterns are limited to `~$*`, `.tmp`, `.temp` to avoid deleting real student project files.

▯ LINUX

▯ System Requirements (Linux)

Requirement	Specification

Requirement	Specification
Supported Distributions	Debian, Ubuntu, Fedora, RHEL, CentOS, Arch, Manjaro, OpenSUSE
Bash	Version 4.0 or higher (pre-installed)
Privileges	root or sudo (mandatory)
Disk Space	50 MB minimum
Package Manager	apt, dnf, yum, pacman or zypper

🐧 Installation on Linux - STEP BY STEP

Step 1: Download or transfer the tool

Option A - If you downloaded on Windows: 1. Copy the `Herramienta-toolbox` folder to your Linux system 2. Use USB, shared network, or FileZilla/SCP

Option B - Download directly on Linux: 1. Open a terminal 2. Download to your home folder: `bash cd ~ # Download or extract the file here`

Step 2: Navigate to Linux folder

1. Open a **Terminal** (Ctrl + Alt + T on most distributions)
2. Navigate to the folder:

   ```
   cd /path/where/is/Herramienta-toolbox/Linux
   ```

   Example:

   ```
   cd ~/Downloads/Herramienta-toolbox/Linux
   ```

3. Verify you're in the right place:

   ```
   ls -l
   ```

   You should see:

   ```
   toolbox.sh
   ```

Step 3: Give execution permissions

⚠ **IMPORTANT**: Scripts need execution permissions

```
chmod +x toolbox.sh
```

Explanation: - `chmod +x` = Give execution permissions - This is only done ONCE

Step 4: Run with sudo

⚠ **IMPORTANT**: You must run with sudo (as root)

Run:

```
sudo ./toolbox.sh
```

What does each part mean? - `sudo` = Run as superuser (root) - `./` = Run from current folder - `toolbox.sh` = Script name

It will ask for your password: Type it and press Enter (You won't see anything while typing, this is normal for security)

Step 5: Select Execution Profile

You'll see this screen:

```
=========================================================================================
                     Renggli PC Solution - SUITE ENTERPRISE V14 (LINUX)
=========================================================================================
Current Log: /path/Logs/Audit_2024-02-10.log
Distribution: ubuntu 22.04 | Package manager: apt

[PROFILE SELECTION]

1. DIAGNOSTICS    - Read-only, audit and queries (no modifications)
2. REPAIR         - Automated maintenance and repairs
3. ADMINISTRATION - Full access (includes formatting and critical operations)

=> Select profile [1-3]:
```

Choose your profile: - **1** = Diagnostics only, makes no changes - **2** = Allows repairs and updates - **3** = Full access (disk formatting, conversions)

Type the number and press **Enter**.

Step 6: Use the Main Menu

You'll see a menu with **30 options** organized in categories:

```
==========================================================================================
   [ HARDWARE DIAGNOSTICS ]         [ SYSTEM REPAIR ]                [ NETWORK & CONNECT
   1. SMART Disk Status             6. Verify System (fsck)          11. Network Reset
   2. Complete Hardware Info        7. Repair Package Manager        12. Speed Test
   3. RAM Memory Test               8. Deep Cleanup                  13. DNS/Ports Audit
   4. OS System Info                9. Repair Bootloader (GRUB)      14. Firewall Diagno
   5. Temperature & Sensors         10. Docker Cleanup               15. Live Network Mo

   [ STORAGE MANAGEMENT ]           [ SERVICES & PROCESSES ]         [ AUTOMATION ]
   16. Secure USB Format            21. Services Management          26. Update System
   17. MBR to GPT Conversion        22. Top CPU/RAM Processes        27. Scheduled Shutd
   18. Disk Analysis                23. View System Logs             28. Data Backup
   19. Partition Mounting           24. Users & Permissions          29. Battery Report
   20. Disk Space                   25. Real-Time Monitoring         30. Verify Integrit

   [0] EXIT WITH REPORT             [00] EXIT WITHOUT REPORT AND WITHOUT LOG
==========================================================================================
```

To use a function: 1. Type the option **number** 2. Press **Enter** 3. Follow the instructions 4. Press **Enter** to continue after each operation

Step 7: Exit

To exit: - Type **0** to generate HTML report and exit - Type **00** to exit without generating report (log + checksum are saved)

About the HTML report: - Option **0** automatically generates an HTML file - The report includes all system information and operation logs

---

▯ Where are the logs? (Linux)

Logs are saved in:

```
Linux/Logs/
├── Audit_2024-02-10.log           (Operations record)
└── Report_Linux_2024-02-10.html   (Visual report)
```

To view the log:

```
cat Logs/Audit_2024-02-10.log
```

To open the HTML report:

```
firefox Logs/Report_Linux_2024-02-10.html
# Or your preferred browser
```

▯▯ Troubleshooting (Linux)

Error: “Permission denied”

Solution:

```
chmod +x toolbox.sh
sudo ./toolbox.sh
```

Error: “No such file or directory”

Solution: Verify you’re in the correct folder:

```
pwd  # Shows current path
ls   # Lists files
```

Error: “This script requires root privileges”

Solution: You must use `sudo`:

```
sudo ./toolbox.sh
```

Dependencies don’t install automatically

Solution: The tool detects and automatically installs packages like: - smartmontools (for SMART) - lm-sensors (for temperatures) - speedtest-cli (for speed test)

If it fails, install manually:

```
# Debian/Ubuntu
sudo apt install smartmontools lm-sensors speedtest-cli

# Fedora/RHEL
sudo dnf install smartmontools lm_sensors speedtest-cli

# Arch
sudo pacman -S smartmontools lm_sensors speedtest-cli
```

□ macOS

□ System Requirements (macOS)

Requirement	Specification
Operating System	macOS 10.14 (Mojave) or higher
Shell	Bash or Zsh (pre-installed)
Privileges	Administrator with sudo (mandatory)
Disk Space	50 MB minimum

Requirement	Specification
Xcode CLI Tools	Auto-installs if missing
Homebrew	Recommended (can auto-install)

⍰ Installation on macOS - STEP BY STEP

Step 1: Download the tool (macOS)

1. Download the `Herramienta-toolbox` folder
2. Extract the file (double-click the .zip)
3. Move the folder to your preferred location (Example: `~/Documents/`)

Step 2: Open Terminal

Method 1 - Spotlight: 1. Press `Cmd + Space` 2. Type "Terminal" 3. Press Enter

Method 2 - Finder: 1. Open **Finder** 2. Go to **Applications** → **Utilities** 3. Double-click **Terminal**

Step 3: Navigate to Mac folder

In the terminal, type:

```
cd /path/where/is/Herramienta-toolbox/Mac
```

Example:

```
cd ~/Documents/Herramienta-toolbox/Mac
```

Trick: You can drag the folder into the terminal after typing `cd` followed by a space.

Verify you're in the right place:

```
ls -l
```

You should see:

```
toolbox.sh
```

Step 4: Give execution permissions

```
chmod +x toolbox.sh
```

Step 5: Allow execution in Security (First time)

⚠ **IMPORTANT FOR macOS:**

macOS blocks scripts downloaded from the internet. To allow them:

Option A - Remove quarantine:

```
xattr -d com.apple.quarantine toolbox.sh
```

Option B - If security message appears: 1. Try to run the script: `bash sudo ./toolbox.sh` 2. If macOS blocks it, you'll see a message 3. Go to **System Preferences** → **Security & Privacy** 4. In the **General** tab, you'll see a message about the blocked script 5. Click **"Allow Anyway"** 6. Run the script again

Step 6: Run with sudo

⚠ **IMPORTANT**: You must run with sudo

Run:

```
sudo ./toolbox.sh
```

It will ask for your macOS password: Type it and press Enter (You won't see anything while typing, this is normal)

Step 7: Select Profile

You'll see this screen:

```
==========================================================================================
                      Renggli PC Solution - SUITE ENTERPRISE V14 (macOS)
==========================================================================================
Current Log: /path/Logs/Audit_2024-02-10.log

[PROFILE SELECTION]

1. DIAGNOSTICS     - Read-only, audit and queries
2. REPAIR          - Maintenance and repairs
3. ADMINISTRATION - Full access (critical operations)

=> Select profile [1-3]:
```

Choose the number and press Enter.

Step 8: Use the Main Menu

The macOS menu currently includes 14 options, organized by profile.

Core macOS functions include: - Hardware, resources, and process diagnostics - Disk and file-system verification - Network diagnostics and reset - System update and cleanup - Scheduled shutdown and system report

Exit options (macOS): - **0** = exit with report - **00** = exit without report and without log

---

? Where are the logs? (macOS)

Logs are saved in:

```
Mac/Logs/
├── Audit_2024-02-10.log
└── Report_Mac_2024-02-10.html
```

To view the log:

```
cat Logs/Audit_2024-02-10.log
```

To open the HTML report:

```
open Logs/Report_Mac_2024-02-10.html
```

?? Troubleshooting (macOS)

Error: “Operation not permitted”

Solution: Run with `sudo`:

```
sudo ./toolbox.sh
```

Error: “command not found: brew”

Solution: Some functions use Homebrew. Install it:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/insta
```

macOS blocks execution

Solution:

```
xattr -d com.apple.quarantine toolbox.sh
```

Or go to **System Preferences → Security & Privacy** and allow the script.

Xcode Command Line Tools not installed

Solution: macOS will ask you to install them automatically. If not:

```
xcode-select --install
```

▯ PROFILE SYSTEM (All Systems)

▯ DIAGNOSTICS (Read-Only)

- ▯ View hardware information
- ▯ Status and queries
- ▯ Does NOT make system changes

Ideal for: - Audits - Preventive checks - When you DON'T have authorization to make changes

▯ REPAIR (Maintenance)

- ▯ Everything from DIAGNOSTICS +
- ▯ Temporary file cleanup
- ▯ Automatic repairs
- ▯ System updates
- ▯ Does NOT allow formatting or conversions

Ideal for: - Daily technical support - Routine maintenance - Solving common problems

▯ ADMINISTRATION (Full Access)

- ▯ EVERYTHING (DIAGNOSTICS + REPAIR) +
- ▯ Disk formatting
- ▯ MBR/GPT conversions
- ▯ Critical operations

▯▯ **CAUTION:** This profile can delete data

Ideal for: - Senior technicians - New equipment preparation - Authorized critical operations

---

▯ FEATURE COMPARISON

Feature	Windows	Linux	macOS
SMART disk status	▯	▯	▯
Hardware Info	▯	▯	▯
RAM Test	▯	▯	▯▯
Repair System	▯ DISM/SFC	▯ fsck	▯ diskutil
Updates	▯ Winget	▯ apt/dnf/pacman	▯ brew/softwareupdate

Feature	Windows	Linux	macOS
Network Reset	□	□	□
Speed Test	□	□	□
Disk Formatting	□	□	□
MBR → GPT	□	□	□
Docker	□	□	□
Firewall	□	□ UFW/firewalld	□
Battery Report	□	□	□

□ NEW WINDOWS MODULES (V14)

The following modules were added to `Windows/toolbox.bat` :

- Critical System Events (disk/power)
- BSOD Analysis (Minidump + Event ID 1001)
- Process Forensics Audit (temp paths + digital signature)
- RAID/Storage Status (Storage cmdlets + WMI fallback)
- Driver Backup (DISM export-driver)
- High Security Profile (Blindaje V1 integrated: persistent T: mapping, strict ACLs, offline Explorer policies, daily folder redirection, and built-in verify/undo)

Notes:

- Diagnostic modules are read-only.
- `Driver Backup` writes files to disk and requires free space.
- `High Security Profile` modifies registry, ACLs and the student hive. After `Undo` , reboot and manually verify in `C:\` whether `Trabajos Alumnos` was fully removed; locked files can leave it behind.
- This module is Windows-only. Linux and macOS keep their own module sets and do not include an equivalent blindage module.

□ □ PROGRAMMER GUIDE: HOW TO ADD NEW MODULES

This section explains how to extend Toolbox on **Windows, Linux, and macOS** in a safe and consistent way.

1) Recommended skills

Before adding modules, it is recommended to have:

- Windows: Batch (`.bat`) and admin commands (`DISM` , `SFC` , `PowerShell` , `netsh` , `wmic` alternatives).
- Linux/macOS: Bash, privilege model (`sudo` , `root`), system tools (`systemctl` , `journalctl` , `ip` , `diskutil` , `launchctl`).
- Operational safety: understanding destructive commands (disk, network, users, shutdown) and impact.
- Auditability: ability to keep clear, timestamped logs.
- Git workflow: branches, small commits, local validation, pull requests.

2) Where to add functions per OS

One edition per operating system:

- Windows: `Windows/toolbox.bat`
- Linux: `Linux/toolbox.sh`
- macOS: `Mac/toolbox.sh`

General rule:

- Add the menu option under the right profile (`DIAGNOSTICS` , `REPAIR` , `ADMINISTRATION`).
- Route the option to a dedicated module (`MOD_...` in Windows, `mod_...` in Linux/macOS).
- Implement the module as an isolated block/function.
- Return to menu cleanly after execution.

3) Minimum structure for a new module

A new module should include:

- Clear title in UI (what action it performs).
- Pre-checks (privileges, command availability, valid input).
- Extra confirmation for sensitive actions.
- Main execution with controlled error handling.
- Log entry with timestamp.
- Clean return to menu.

Naming conventions:

- Variables: `UPPER_SNAKE_CASE` .
- Modules: prefix `MOD_` (Batch) or `mod_` (Bash).
- Validation helpers: prefix `CHECK_` / `check_` .

4) Mandatory safety requirements

If the module changes system, disk, or network state, you must:

- Never target system disk/partition without explicit block logic.
- Avoid broad wildcard deletion outside Toolbox-owned scope.
- Avoid modifying external scheduler jobs not owned by Toolbox.
- Keep admin/root checks intact.

- Show explicit warnings before irreversible actions.

5) Platform-specific considerations

- Windows:
  - Use commands compatible with supported Windows versions.
  - Validate external tools before calling them.
  - Keep report/checksum behavior consistent.
- Linux:
  - Handle distro differences (apt, dnf, yum, pacman, zypper).
  - Avoid hardcoded assumptions for paths/services.
  - Keep pipelines resilient to non-critical command misses.
- macOS:
  - Validate support on Intel and Apple Silicon.
  - Do not assume non-default dependencies are installed.
  - If using `launchd`, scope changes to Toolbox identifiers only.

6) Quick pre-release checklist

1. Module appears only in the correct profile(s).
2. Includes validation, confirmations, and logging.
3. Does not break exit options (`0` and `00`) or menu return.
4. Introduces no unsafe cleanup/shutdown/disk patterns.
5. Basic manual validation was completed on target OS.

7) Documentation updates required

When behavior changes or a module is added, update:

- `HISTORIAL_DE_CAMBIOS.md`
- `Manuales/README_ES.md`
- `Manuales/README_EN.md`
- `Manuales/README_CN.md`

Also follow the contribution workflow defined in:

- `CONTRIBUTING.md`

8) Recommended implementation flow

1. Define module goal and target profile.
2. Implement in the script for the target OS.
3. Test success and failure paths.
4. Verify logs, exit behavior, and menu return.
5. Update docs and changelog.
6. Open PR with scope, risks, and validation evidence.

DETAILED MENU AND OPTION CATALOG

To understand **each option** (what it does, why it exists, when to use it, and precautions), see:

- `Manuales/CATALOGO_OPCIONES_EN.md`

Includes coverage for:

- Windows, Linux and macOS
- Profiles `DIAGNOSTICS`, `REPAIR`, `ADMINISTRATION`
- Risk labels `[R]`, `[W]`, `[!]`

Robustness Note (Update 16)

- Windows, Linux, and macOS now validate shutdown schedule input using `HH:MM` format.
- HTML reports now escape special log characters (`&`, `<`, `>`) to prevent rendering issues.
- SHA256 checksum is now saved to a sidecar `*.sha256` file to avoid self-invalidating the log hash.

SUPPORT

Problems or questions? - Email: tomasrenggli@gmail.com - Complete documentation in `Manuales/` folder

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VERSION

Toolbox V14 Multi-Platform - Windows: 23 modules - Linux: 30 modules - macOS: 14 modules